COURSE OUTLINE

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Agentic Virtual Patching Platform

From vulnerability finding to tested interim mitigation

- 3 days (8 hours per day)
- Intermediate to advanced
- Instructor-led, hands-on labs
- Capstone project

About this course

Who should attend

Application security, product security, DevSecOps, WAF and platform engineers, vulnerability management teams.

Prerequisites

- Basic HTTP and web application security
- Reverse proxy and WAF concepts
- Python and container basics
- Familiarity with an AI coding assistant is helpful, not required

✓ What you will be able to do

- Explain the role and limits of virtual patching as an interim mitigation, not a substitute for a permanent fix
- Build a finding intake and evidence and context model with provenance
- Generate candidate rules for a lab enforcement point using constrained templates and strict schemas
- Create positive, negative, regression and bypass-oriented test cases and fail closed on incomplete validation
- Produce an approval package with test evidence, rollback instructions and monitoring criteria

Tools and technologies

- Claude
- Python
- FastAPI
- Pydantic
- pytest
- Docker Compose
- NGINX / ModSecurity lab
- OWASP
- SQLite

GitHub

Table of contents

DAY 1 3 modules

MODULE 1

Virtual patching fundamentals and the agentic baseline

- What a virtual patch can and cannot do; interim mitigation versus permanent fix
- Threat model, HTTP and WAF basics, the lab enforcement point and deliberately vulnerable endpoint
- Shared engineering principles: model output as a proposal, authorization outside the model, human approval
- The AI-assisted development workflow used in every build

HANDS-ON LAB
Stand up the lab with Docker Compose and exercise the vulnerable endpoint behind the proxy.

DELIVERABLE
Reproducible lab and product brief.

MODULE 2

Finding ingestion and evidence model

- Typed vulnerability record: advisory ID, affected component and endpoint, evidence references, confidence, source timestamp
- Mock scanner JSON importer with schema validation and provenance
- Invalid-input cases and rejected-row reporting

HANDS-ON LAB
Implement the finding normalizer and unit tests with the AI coding assistant; inspect every diff.

DELIVERABLE
Working finding intake.

MODULE 3

Context collection

- Endpoint, application owner, environment and route metadata from fixtures or the demo app
- Never inferring reachability without evidence
- Marking unknowns explicitly

HANDS-ON LAB
Build the context collector and attach it to each finding.

DELIVERABLE
Finding plus context bundle.

DAY 2

3 modules

MODULE 4

Agentic mitigation planning

- An agent summarizes the issue and proposes a strategy from an approved playbook
- Citations to supplied evidence are mandatory; unknowns are marked
- Structured outputs, evidence-grounded reasoning, model adapter layer

HANDS-ON LAB
Write the planning prompt and output schema; test it with missing and contradictory evidence.

DELIVERABLE
Mitigation planning agent.

MODULE 5

Constrained candidate rule generation

- Rules generated only through approved, versioned templates and a schema
- Rejecting arbitrary directives and anything outside the lab's supported format
- Rule selection logic and its tests

HANDS-ON LAB
Generate candidate rules for the lab finding and prove the generator refuses out-of-scope rules.

DELIVERABLE
Template-constrained rule generator.

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MODULE 6

Test harness and validation gate

- Positive tests for the target case, benign traffic tests, regression tests, bypass variations
- Running against the isolated proxy or WAF and scoring blocked, allowed and limitations
- Fail closed when validation is incomplete; confidence reporting

HANDS-ON LAB
Build the harness, run it, and break a rule with a bypass variant to see the gate fail closed.

DELIVERABLE
Automated validation report.

DAY 3

3 modules

MODULE 7

Approval package and staging deployment

- Review bundle: diff, test results, owner, expiry, rollback command, approver
- Deploy only to lab staging after explicit approval; the deployment adapter
- Expiry and re-validation of interim mitigations

HANDS-ON LAB
Produce the approval package and deploy to staging through the gated adapter.

DELIVERABLE
Approval and deployment workflow.

MODULE 8

Monitoring, rollback and operations

- Request and block metrics from synthetic traffic
- Rollback exercise and operator runbook
- Audit events, versioned artifacts, known limitations

HANDS-ON LAB
Run the rollback drill and write the runbook.

DELIVERABLE
Monitoring and rollback runbook.

MODULE 9

Capstone: end-to-end virtual patching lab platform

- Finding in, validated mitigation out, with no deployment without approval
- Acceptance criteria review, safety and limitations statement, demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end platform and runbook.

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Capstone

Capstone project

A repeatable lab platform that ingests a finding, proposes an evidence-linked mitigation, validates it against target and benign traffic, and ships an approval package with tested rollback.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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